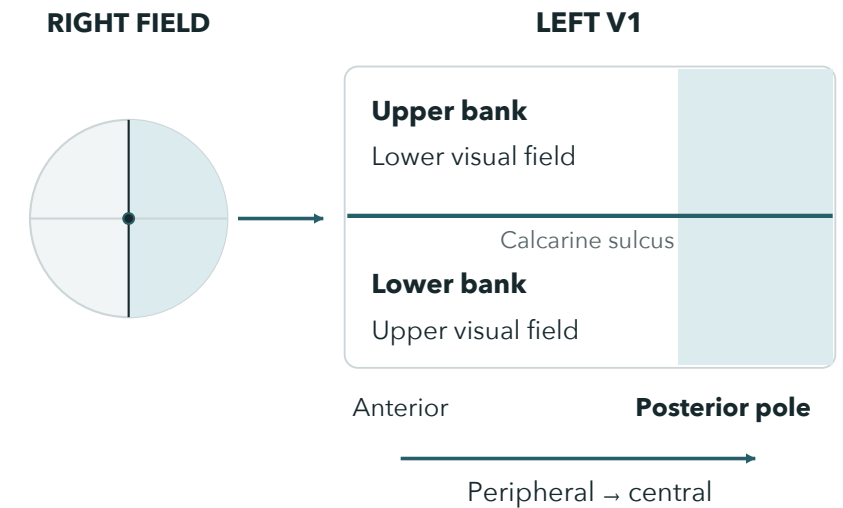


Visual cortex retinotopy

Map side, vertical field and eccentricity before naming a lesion.

Three coordinates on calcarine cortex

A | Left V1 represents the right field



Central field uses more cortex

Superior bank = inferior visual field. Inferior bank = superior visual field.
Posterior pole = central field.

Map before localising

Coordinate	Cortical map
Right visual hemifield	Left primary visual cortex
Lower visual field	Upper calcarine bank
Upper visual field	Lower calcarine bank
Central to peripheral	Occipital pole to more anterior calcarine cortex

Fast check

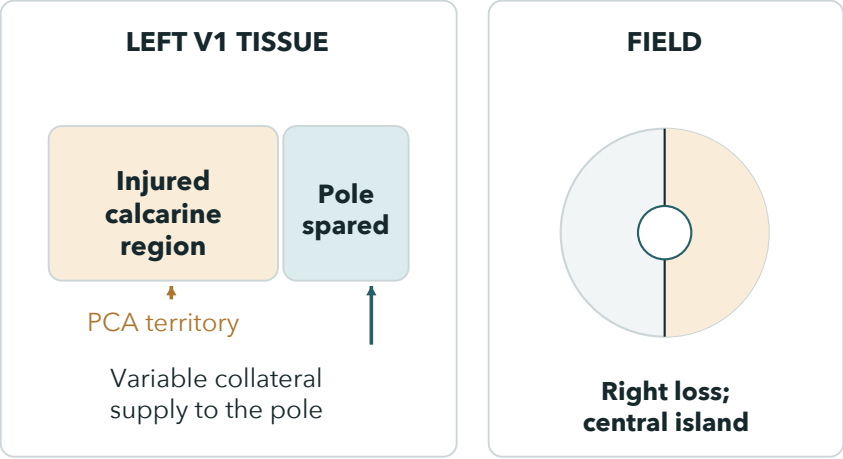
Central vision occupies disproportionately much cortex. A small posterior lesion can impair a central homonymous region while leaving more peripheral eccentricities represented anteriorly.

Figure reference

A: Neuroscience retinotopic representation. Original schematic.

Why the centre may survive

B | Tissue preservation explains a central island



Preserved central cortex → preserved central field

Illustrative left occipital injury produces a right homonymous defect. Collateral supply to the pole is variable.

Use the clue carefully

Observation	Interpretation
Central island in a homonymous defect	Macular-representing tissue may be spared
Occipital pole perfused despite nearby injury	Collateral supply is one possible mechanism
Closely matching incomplete defects	Often favours a posterior site; not an exact-site rule

Fast check

Macular sparing is neither universal in occipital injury nor exclusive to it. Combine the field, imaging and examination instead of treating one pattern as proof.

Figure reference

B: Bookshelf homonymous hemianopsia; V1 retinotopy. Original schematic.